TOPIC 03 · INTERACTIVE DASHBOARDS WITH COPILOT, SKILLS AND RULES
LAB 8 THEME · DASHBOARD SKILL, REFRESH & SHARE

Lab 8 — Turn the Dashboard into a Reusable Skill, Refresh It and Share It Safely

Maps to	LO3: use Copilot to write a dashboard skill, run it on a workforce workbook with rules, prove the dashboard refreshes with new rows, and check it is safe to share
Duration	45 minutes
Tools	Microsoft 365 Copilot web (https://copilot.cloud.microsoft/) to draft the skill · Excel Copilot to run it · .Rules · PivotTables, slicers, refresh
Mock data	data/BrightPath_Workforce_Headcount.xlsx
Lab files	rules-template.txt (the .Rules text to paste into column A); skills/hr-workforce-dashboard/SKILL.md (ready-made skill to upload if you run short of time)
Course	Microsoft 365 Copilot for Excel (C34) · v3.1
Prompts	https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-8

Goal

Build BrightPath's workforce and attrition dashboard from a one-line skill call, prove it updates when new joiners arrive, and make sure no small group can be identified.

What You Will Do

BrightPath's HR Director needs the same kind of dashboard every month. Instead of repeating a long brief, you will ask Microsoft 365 Copilot at https://copilot.cloud.microsoft/ to write an hr-workforce-dashboard skill, save it to OneDrive and return to Excel to run it on the workforce workbook, whose .Rules sheet holds the period, the attrition definition and the privacy rule. Then you will append eight July joiners and refresh to prove the dashboard is live, check that no slicer combination exposes fewer than 5 people, and ask Copilot for a short leadership summary.

What You Will Build

An hr-workforce-dashboard skill in OneDrive, and BrightPath_Workforce_Headcount.xlsx with a .Rules sheet, helper columns, a Checks sheet and an HR_Dashboard with six KPI cards, four charts and Location and Job_Level slicers - refresh-tested and privacy-checked

Prerequisites

- Lab 6 and Lab 7 completed: you can save a custom skill to OneDrive and build a dashboard with rules.
- This lab folder's data workbook, rules-template.txt and skills/hr-workforce-dashboard/SKILL.md (fallback).
- Access to https://copilot.cloud.microsoft/ with the same work account used in Excel and OneDrive.

Data note

Use only the supplied fictional BrightPath data or information you are authorised to process. Do not paste credentials, confidential business material or real employees' personal data into Copilot, rules or skills.

Steps

Step 1. Copy BrightPath_Workforce_Headcount.xlsx from this lab's data folder into a work-c34 folder on your computer and open it in Excel for Microsoft 365 (desktop). Work on this local copy - no OneDrive upload is needed. Check that Formulas > Calculation Options is Automatic, then open Copilot from the Copilot button on the Home tab or the floating Copilot icon at the bottom right. Read the About_This_Data sheet before you prompt. Do not touch the New_Joiners_Jul2026 sheet until the refresh test.

PROMPT / CHECK — copy and adapt

```
Table : tblEmployees (420 employees hired since 2019; Status Active or Exited)
Period: 2025-07-01 to 2026-06-30
Attrition rate = exits in period / ((active on 2025-07-01 + active on 2026-06-30) / 2)
```

Step 2. Create the workbook rules: + (Add work content) > Create workbook rules, then paste rules-template.txt into .Rules!A1 or send the prompt below. Note the privacy rule - the skill will follow it.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
On the .Rules sheet, replace everything in column A with these rules, one per cell from A1
down.
Keep each ## heading in its own cell. Do not change any other sheet.
## WORKFORCE DASHBOARD STANDARDS
Never change or delete existing rows of tblEmployees.
Never show Employee_ID, Monthly_Salary or any individual record on a dashboard.
The 12-month period is 2025-07-01 to 2026-06-30. Active on a date = Hire_Date on or before it
and Exit_Date blank or after it.
Active headcount = employees with Status Active now, so new joiners count as soon as they are
added to tblEmployees.
Attrition rate = exits in the period / ((active on 2025-07-01 + active on 2026-06-30) / 2).
New joiners after 2026-06-30 do not change it.
Tenure_Years = years from Hire_Date to Exit_Date, or to 2026-06-30 if still active; use 0 for
anyone hired after 2026-06-30.
Do not show any group of fewer than 5 employees. Group it into Other or leave it out.
## LAYOUT
Dashboard sheet HR_Dashboard: title in A1, "Data as at 2026-06-30" in A2, KPI cards in row 4,
charts in a grid below, slicers on the right.
Put every PivotTable on a sheet named Dash_Data. Never merge cells. Hide gridlines on
HR_Dashboard.
## VISUAL STYLE
Colours: navy #1F4E79 for headcount, teal #2A9D8F for hires, amber #F4A261 for exits and
attrition.
Every chart title states the finding. Rates as % with 1 decimal. Years with 1 decimal.
```

Step 3. Open https://copilot.cloud.microsoft/ in your browser and sign in with the same work account used in Excel. Start a new chat in Microsoft 365 Copilot on the web and paste Prompt B below to draft the skill. The web chat does not automatically see your local workbook or Excel chat, so this prompt supplies the table schema and dashboard recipe. No workbook upload is needed. Read the returned text: it must start with --- and include name: hr-workforce-dashboard and a description, followed by the numbered steps. Ask for corrections in this web chat before saving.

PROMPT / CHECK — copy and adapt

```
PROMPT (Microsoft 365 Copilot web):
Write a complete SKILL.md for a Copilot in Excel skill named hr-workforce-dashboard.
Return reusable skill instructions; do not build a dashboard or claim to inspect my local
workbook in this chat.
Start with YAML front matter containing name: hr-workforce-dashboard and a description.
Description: Use when I ask for a workforce, headcount or attrition dashboard from an employee
table.
Workbook context for the skill when it runs in Excel: tblEmployees has Employee_ID,
Department, Location, Job_Level, Employment_Type, Hire_Date, Exit_Date, Exit_Type, Status,
Monthly_Salary, Engagement_Score, and Training_Hours_YTD. Status is Active or Exited; blank
Exit_Date means no recorded exit.
```

```
The .Rules worksheet holds the reporting period (2025-07-01 to 2026-06-30 for this lab), KPI definitions, privacy limits and formatting. Read it at run time; ask if it is missing or unclear.
Attrition = exits in period / ((headcount at period start + headcount at period end) / 2).
Voluntary attrition uses voluntary exits and the same denominator; handle a zero denominator explicitly.
Never display Employee_ID, salary, individual records or groups smaller than 5 on the dashboard.
Steps:
1) Read the .Rules sheet first and follow it.
2) Add helper columns Active_Start, Exit_In_Period and Tenure_Years using the period in the rules.
3) Create a Checks sheet with independent COUNTIFS formulas for active headcount (Status Active), headcount at period start, headcount at period end, exits, voluntary exits, attrition rate and hires.
4) Put all PivotTables on Dash_Data.
5) Build HR_Dashboard: six KPI cards (Active headcount, Exits, Attrition rate, Voluntary attrition rate,
   Avg tenure of active staff, Hires), charts of headcount by Department, attrition rate by Department, hires vs exits by quarter and headcount by Location, and slicers for Location and Job_Level.
6) Compare every KPI card with Checks and report any difference.
7) Check slicer combinations for groups smaller than 5; suppress their values and report the privacy check.
8) Use tblEmployees as a dynamic source and refresh all PivotTables after new rows are appended.
Return only the full SKILL.md text, starting with the --- front matter, without code fences or commentary.
```

Step 4. Copy the skill text from the web chat into a plain-text file named SKILL.md (not SKILL.md.txt). Do not include Markdown code fences around the file. Return to Excel: Settings (...) > Manage plugins & skills > Custom skills > Open skills folder (select Create OneDrive folder first if needed). Create the folder hr-workforce-dashboard in your OneDrive skills folder and upload the file (or upload the ready-made one from this lab folder). In Excel select Settings (...) > Manage plugins & skills > Custom skills > Refresh and turn the skill on.

PROMPT / CHECK — copy and adapt

```
OneDrive : .../skills/hr-workforce-dashboard/SKILL.md
Check : folder name = name: line = hr-workforce-dashboard
```

Step 5. Run the skill in Plan mode: one line replaces the long brief, because the skill holds the steps and the rules hold the definitions. Review the plan, approve it, and select Done when Copilot finishes.

PROMPT / CHECK — copy and adapt

```
PLAN MODE:
@hr-workforce-dashboard Build the workforce dashboard for the 12 months ending 2026-06-30.
```

Step 6. Verify. Ask Copilot to compare the KPI cards with the Checks sheet the skill built, and to show the attrition formula. Compare both with the Test It values - a skill can build a wrong check too.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Compare each KPI card on HR_Dashboard with the Checks sheet and tell me which, if any, do not match.
Show the formula behind the Attrition rate card.

ALLOW EDITING (if needed):
The attrition rate must use average headcount = (active on 2025-07-01 + active on 2026-06-30) / 2, as the rules say.
```

Step 7. Refresh test: ask Copilot to append the 8 rows from New_Joiners_Jul2026 to tblEmployees and refresh every PivotTable. Every KPI and chart that counts active staff must update.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
Copy the 8 data rows in New_Joiners_Jul2026!A2:L9 to the first empty rows directly below
tblEmployees so the table expands. Then refresh all PivotTables in the workbook and tell me
the new Active headcount and the number of rows in tblEmployees.

Expected after refresh:
Active headcount : 237 -> 245
tblEmployees rows: 420 -> 428
New joiners by department: Recruitment +2, Sales & Client Services +2, IT +4
```

Step 8. Check it is safe to share, then ask for a leadership summary that uses only dashboard numbers. Copilot drafts; you check every number before it leaves HR. Manually fill in the Step 3 / Prompt B row in Prompt_Log: record that the web chat drafted SKILL.md and how you checked the front matter and steps. Excel Copilot cannot see that web conversation. Then log the Excel prompts with Prompt I.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Review HR_Dashboard before I share it with the leadership team. Does any element expose
individual employee data or a group smaller than 5? Is each KPI labelled with its period and
definition?

CHAT ONLY:
Write a 5-line summary of HR_Dashboard for BrightPath's leadership team. Use only numbers
shown on the dashboard, and label any explanation as a hypothesis to test.

LOG (Allow editing):
In Prompt_Log, leave the Step 3 / Prompt B web-chat row unchanged. For each Excel prompt I
ran, fill in What Copilot did and How I verified, and set Decision to Keep. Do not invent
results for web prompts.
I will tell you any row to change to Revise or Undo.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online:
https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-8

#	Purpose	Prompt
1	Write a skill · Microsoft 365 Copilot web	In the web chat at https://copilot.cloud.microsoft/, use the complete Step 3 prompt below to draft hr-workforce-dashboard. It includes the workbook schema and rules that this chat cannot see.
2	Run a skill	@hr-workforce-dashboard Build the workforce dashboard for the 12 months ending 2026-06-30.
3	Refresh	I added rows to tblEmployees. Refresh all PivotTables and tell me which KPI values changed.
4	Privacy	Which slicer combinations show a group of fewer than 5 employees? Suggest how to suppress them.
5	Explain	Explain how the Attrition rate card is calculated, step by step.
6	Improve the skill · Microsoft 365 Copilot web	I will paste my current hr-workforce-dashboard SKILL.md below. Update it so it also adds a timeline on Hire_Date. Preserve its front matter, KPI definitions and privacy rules. Return the full file without code fences. If I have not pasted the file, ask me for it first.
7	Summary	Write a 5-line leadership summary using only the numbers on HR_Dashboard.

Test It

Check	Expected result
Active headcount	237 before the refresh
Exits and attrition	29 exits (25 voluntary); 13.0% attrition
Hires and tenure	56 hires; average tenure about 3.1 years
Highest-attrition department	Recruitment: 21.8% (12 exits)
Refresh test	After adding the July joiners: headcount 245, 428 rows
Privacy	No Employee_ID, salary, individual record or group under 5 on the dashboard

Checkpoint

You now have two custom skills in OneDrive. Run hr-workforce-dashboard again next month on new data, and improve the skill whenever you correct Copilot twice.

Troubleshooting

If this happens	Fix
The web chat asks to see my workbook or outputs a dashboard instead of SKILL.md.	Use the full Step 3 prompt at https://copilot.cloud.microsoft/. Ask for reusable skill instructions only; the local workbook is used later in Excel. Do not upload it just to draft the skill.
The skill does not appear in Excel after saving.	Use the skills folder opened from Excel, check the file is SKILL.md rather than SKILL.md.txt, and ensure the folder name and front-matter name are both hr-workforce-dashboard. Select Refresh and enable it. If custom skills are unavailable for your account, use the supplied skill text as a normal Excel prompt and record that the installation could not be completed.
The new rows did not join the table and nothing changed on refresh.	Ask Copilot to resize tblEmployees to include the 8 new rows, then refresh all PivotTables again.
Attrition shows 12.2% (29 / 237) instead of 13.0%.	The card divides by closing headcount only. Ask Copilot to use the average of opening and closing headcount, then add that wording to the skill.
The skill ignores the .Rules sheet.	Check the sheet is named .Rules and is visible, and that step 1 of the skill says to read it. Start a New chat.
Helper columns show #VALUE! for blank Exit_Date.	Ask Copilot to treat blank Exit_Date explicitly, e.g. OR([@Exit_Date]="",[@Exit_Date]>DATE(2025,7,1)).

Challenge

Improve the skill so it also adds a Timeline on Hire_Date and a chart of average Engagement_Score by Department for active staff. Save, refresh and rerun it, then check one department's average with AVERAGEIFS.

Reflection

What would happen to trust in this dashboard if one leader could filter it down to a single named team member?